Time : 3 hours **Max. Marks :**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

Marks

1. a) Design a NFA which accepts the set of all strings that start with zero. 2

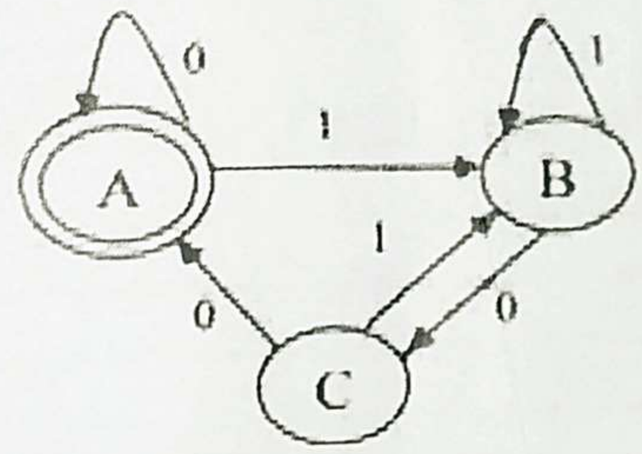
b) Define ambiguous grammar and CFG. 2

c) What are the Properties of CFLs. 2

d) What is the class of language for which the TM has both accepting and rejecting configuration? Can this be called a Context free Language? Discuss. 2

e) What is a recursively enumerable language and recursive sets? 2

f) Construct regular expression corresponding to the state diagram: 2



2. a) What is regular expression? Explain different properties of regular expressions and design regular expressions for following languages: 6

(i) L= {w: w ε (a + b)*, w has odd number of a's }

(ii) L= {$a^n b^m$ | n≥ |3|, m ≤2}

b) Differentiate between Mealy and Moore machine. Construct a Moore machine to compute residue modulo for base 4 (0,1,2,3) and convert it to Equivalent Mealy machine. 6

3. a) Explain about derivation and parse trees? Construct the string 0100110 from the Leftmost and Rightmost derivation. 6 3

{P.T.O.}

S→0S/1AA
A→0/1A/0B
B→1/0BB

b) Give the new set of productions after removing the unit productions for the following CFG. 6 2

$S \rightarrow AA,\ A \rightarrow B \mid BB,\ B \rightarrow abB \mid b \mid bb.$

4. a) What is LR(0) grammar and explain how LR(0) parsers work with an example. 6

b) Construct PDA accepting $\{ a^n b^m a^n \mid m, n \geq 1 \}$ by final state. 6

5. a) Design a Turing machine which computes the function $f(n) = 3$ if $n \geq 5$ and $f(n) = 0$ if $n = 0, 1, 2, 3$ or 4. 6

b) Define a multi-headed Turing machine, a model in which each tape can have k tape heads. Prove that a Deterministic Turing Machine (DTM) with one work tape can simulate a two-headed Turing machine. 6

6. a) Prove that the union of two recursively enumerable languages is recursively enumerable. 5

b) Discuss post correspondence problem. Let $\sum = \{0,1\}$. Let A and B be the lists of three strings each, defined as 7

	A	B
i	wi	xi
1	1	111
2	10111	10
3	10	0

(i) Does the PCP have a solution

(ii) Prove that the universal language is recursively enumerable

7. a) Write down procedure for Myhill- Nerode theorem with a given example. ('*' means final states). 6

Present State	Next State	
	I/P=a	I/P=b
→ A	B	F
B	A	F
C	G	A
D	H	B
E	A	G
*F	H	C
*G	A	D
*H	A	C

b) Convert the grammar

S →aAA

A →aS/bS /a

to a PDA that accepts the same language by empty stack.

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